

Treasury Products — Quick Reference Card

Universal pricing recipe · 33 products · risk metrics · ALM patterns. Deep companion: treasury_products_cheatsheet.pdf

◆ Universal Symbols

N	notional (domestic)
T, τ	maturity / year fraction
K	strike / contracted rate (LOCKED)
DF(t)	$\exp(-z(t) \cdot t)$
DF(t,s)	$\exp(-(z(t)+s/1e4) \cdot t)$
S, fx	FX spot (dom/foreign)
b	breakeven inflation
A(T)	$\sum \tau \cdot DF(t_i)$ (annuity)
σ , N($\cdot$)	vol, std normal CDF
Master skeleton: $V = (\text{market_implied} - \text{contract}) \cdot \text{scale} \cdot \text{discount}$	

◆ Master Recipe (4 steps)

1. **List cashflows** — when, ccy, contingent on what
2. **Substitute each random CF** with risk-neutral expectation ($\downarrow$)
3. **Discount** by DF(t) (or DF(t,s) for credit)
4. **Sum**: $V = \sum E[CF_t] \cdot DF(t)$

SUBSTITUTION TABLE

Random CF	Replace with
floating rate L	$(D_1/D_2 - 1)/\tau$
spot FX at T	$S \cdot DF_f/DF_d$
inflation index	$(1+b)^T$
option max($S-K, 0$)	$DF(T) \cdot [F \cdot N(d_1) - K \cdot N(d_2)]$
survival to t	$Q(t) = \exp(-\lambda t)$
default in $(t_1, t_2]$	$Q(t_1) - Q(t_2)$
swap rate (CMS)	fwd + convexity adj.

◆ Universal Patterns

A — LINEAR-IN-STRIKE

$V = (\text{market} - \text{contract}) \cdot \text{scale} \cdot \text{disc}$
Par strike = market-implied (set $V=0$).
Used by: FX fwd, FRA, ZCIS, IRS, CDS

B — BASIS × ANNUITY

$V = b \cdot N \cdot A(T)$ [$\cdot$ FX]
Used by: Sec lending, tenor basis, XCCY basis

C — CASHFLOW STRIP PV

$V = \pm \sum cf_i \cdot \text{uplift } i \cdot DF(t_i, s)$
Used by: Bond, FB, Linker, Loan, Annuity Liab

Recognise the pattern → recall the formula. Each product is one of these three shapes.

◆ Walk-through: FRA via recipe

1. CF: $N(L-K)\tau$ at T_2
2. $L \rightarrow (D_1/D_2-1)/\tau = \text{fwd}$
3. discount: $\cdot DF(T_2)$
4. $V = N(\text{fwd}-K)\tau \cdot DF(T_2)$

Every entry below is a 1-line output of the same recipe.

◆ Reference Curve (toy examples)

t	z(t)	DF(t)
1y	4.50%	0.95600
2y	4.40%	0.91576
3y	4.30%	0.87897
5y	4.20%	0.81058
10y	4.10%	0.66365

$A(5y) = \sum \tau \cdot DF(1..5) = 4.40498$

◆ Portfolio Recipe (ALM)

1. **Exposure vector** — bucket by KRD (2y/5y/10y/30y), IE01, CS01, FX Δ , vega
2. **Aggregate** signed contributions (assets +, liabs −)
3. **Hedging set** — IRS strip + Linker strip + nominal bonds + OIS + ZCIS
4. **Solve linear system** → hedge notionals; report residuals

Parallel DV01 $\neq$ KRD. A 10y liab hedged with 5y IRS is parallel-neutral but long-end exposed. KRD-mismatch = canonical pension failure mode.

◆ Convention reminders

- bp → decimal: $/ 1e4$. 15bp = 0.0015.
- Inflation compounds: $(1+b)^t$ (NOT $1+b \cdot t$).
- τ must match contract day-count.
- Annuity liability is NEGATIVE.
- FRA fwd: SIMPLE $(D_1/D_2-1)/\tau$, not continuous.
- Swap float telescope: $1 - DF(T)$ (single-curve).
- CDS hazard: $\lambda \approx s/(1-R)$; $Q(t)=\exp(-\lambda t)$.
- Black option vol: lognormal in **F**, not **S**.
- Multi-curve (post-2008): OIS discount + LIBOR/SOFR project — drills ignore.

● I — Cash & Money Market

FX Spot (§1)

$V = N \text{ for } \cdot S \cdot \text{fx_factor}$
Own foreign currency. Pure FX delta; no rate or time.

Risk: FX

Repo / Reverse Repo (§2)

$V = N(1 + r \cdot T) \cdot DF(T) - N$
Secured short-term borrow (repo) / lend (reverse). $V \approx 0$ at par; small DV01.

Risk: rate (small)

Sec Lending (§3)

$V = (\text{fee_bp}/1e4) \cdot N \cdot A(T)$

Lend securities; receive fee per period. Pattern B.

Risk: rate (tiny)

Amortising Loan (§4, linear amort)

$V = \sum cf_i \cdot DF(i, \text{spread_bp})$
 $\text{amort} = P/n$, $cf_i = \text{amort} + c \cdot \text{outstanding}$
Equal principal/period + interest on remainder. For annuity-style (equal total pmt) use $pmt = P \cdot c / (1 - (1+c)^{-n})$.

Risk: rate, credit spread (CS01)

E1 T-Bill / discount paper

$V = \text{face} \cdot DF(T)$ (zero coupon, govt)

$dy = (\text{face}-P)/\text{face} \cdot 360/\text{days}$

Risk: rate

E2 Commercial Paper

$V = \text{face} \cdot DF(T, \text{spread_bp})$

Short corp/bank discount paper.

Risk: rate, credit

▲ II — Forwards

FX Forward (§5)

$V = N \text{ for } \cdot (S \cdot \text{fx_factor} - K) \cdot DF(T)$

Locked-rate currency exchange at T. $K_par = S \cdot DF_f/DF_d$ (IRP).

Risk: FX, rate

FRA (§6)

$\text{fwd} = (DF(T_1)/DF(T_2) - 1)/\tau$

$V = N \cdot (\text{fwd} - K) \cdot \tau \cdot DF(T_2)$

Locked rate on future deposit $[T_1, T_2]$. $K_par = \text{fwd}$.

Risk: rate

MTM uses curve-implied fwd; real settle = $N(L-K)\tau/(1+L \cdot \tau)$ paid at T_1 .

■ III — Bonds (Pattern C — cashflow strips)

Vanilla Bond (§7)

$V = \sum cf_i \cdot DF(t_i, \text{spread_bp})$

Coupons + principal, discounted at issuer's credit curve.

Risk: rate (DV01), credit (CS01)

Foreign Bond (§8)

$V = (\sum cf_i \cdot DF(t_i, s_bp)) \cdot S \cdot \text{fx_factor}$

Bond in foreign currency, translated at spot.

Risk: rate, credit, FX

Linker (inflation-linked) (§9)

$V = \sum cf_real_i \cdot (1+b)^t \cdot DF(t_i)$

CFs uplifted by inflation. Long inflation = $b \uparrow \rightarrow V \uparrow$.

Risk: rate, breakeven (IE01)

Indexation lag: 3M (Gilts) / 8M (TIPS) — adds small IE01 basis.

E3 Zero-Coupon Bond (long)

$V = \text{face} \cdot DF(T, s)$

$D_mac = T$; $D_mod = T / (1 + y/m)$

No coupons; max duration per \$ of face.

E4 Floating-Rate Note (FRN) (at reset)

V (at fixing) $\approx \text{face} \cdot \text{spread} \cdot A + \text{face}$

Resets to $\sim \text{par}$; duration $\approx \text{time-to-next-fixing}$. Between resets the locked coupon adds accrual.

Risk: spread (dominant), tiny rate

E5 Callable Bond

$V = V_straight - V_call_option$

Issuer's right to redeem early → bondholder bears asymmetric risk; **negative convexity**.

Risk: rate, credit, vol

E6 Convertible Bond

$V = V_straight + V_equity_call$

Bond + long call on issuer's equity. Multi-factor.

Risk: rate, credit, equity, equity vol

◆ IV — Swaps

Interest Rate Swap (IRS) (§10)

$V_rec = N \cdot [c \cdot A(T) - (1 - DF(T))]$

where $A(T) = \sum \tau \cdot DF(t_i)$

$c_par = (1 - DF(T)) / A(T)$

Exchange fixed-c vs floating; receive-fix sign shown. Pay-fix flips sign.

Risk: rate (THE rates instrument)

ZCIS Inflation Swap (§11)

$V = N \cdot DF(T) \cdot [(1+b)^T - (1+K)^T]$

$K_par = b$

Single payment at T: inflation index vs fixed.

Risk: rate, breakeven

Tenor Basis Swap (§12)

$V = b \cdot N \cdot A(T)$

$b = (\text{basis_bp} + \text{shock_bp})/1e4$

e.g. 3M vs 6M floating spread. Pattern B.

Risk: rate, tenor-basis

XCCY Basis Swap (§13)

$V = N \text{ for } \cdot S \cdot \text{fx_factor} \cdot b \cdot A(T)$

Cross-currency basis leg in foreign ccy; FX sensitive.

Risk: rate, xccy-basis, FX

Real XCCY: principal exchange + MtM resets (drill omits).

Total Return Swap (TRS) (§14)

$V = PV(\text{reference asset}) - \text{funding notional}$

Synthetic long the reference (balance-sheet light).

Risk: inherits reference's risks

E7 Overnight Index Swap (OIS)

Same as IRS; float leg = compounded 0/N

Risk-free swap; benchmark discount curve post-2008.

E8 CMS

float leg = swap rate (e.g. 10y) → convexity adj. needed (SABR surface dependent)

Bet on curve shape (steepening/flattening).

E9 Asset Swap (ASW)

Bond + offsetting IRS → synthetic floater

ASW spread $\approx$ bond coupon − par swap rate

Isolates credit spread, neutralises rate.

E10 YoY Inflation Swap

Each year: $N \cdot (\pi_i - K) \cdot \tau$

(vs ZCIS = single compounded payment)

Hedge year-by-year inflation; convexity adj vs ZCIS.

▲ V — Credit

Credit Default Swap (CDS) (§15)

$\lambda = \text{market_spread} / (1-R)$

$Q(t) = \exp(-\lambda \cdot t)$

$\text{premium} = \sum \tau \cdot DF \cdot Q$

$\text{protect} = \sum (Q_prev - Q) \cdot DF$

$V_buyer = N \cdot [\text{protect} \cdot (1-R) - \text{premium} \cdot K]$

Buyer pays K, receives $(1-R) \cdot N$ on default. Buyer wins when spread widens.

$K_par = \text{market_spread}$.

Risk: credit spread (dominant), rate (small)

E13 CDS Index (CDX/iTraxx)

Same as §15. Each defaulting name pays $(1/n) \cdot (1-R) \cdot N$.

Remaining notional continues to accrue premium.

CDX IG = 125 US IG names @ 100bp; HY @ 500bp. NOT a first-to-default basket.
Risk: spread (CS01), correlation (tranches)
E14 MBS — agency pass-through
| $V = \sum E[CF_t | \text{prepay_model}] \cdot DF(t, OAS)$
| (MC over rate paths + CPR/PSA prepayments)
Negative convexity from prepayment option.
Risk: rate (effective DV01), prepay, OAS, vol

● **VI — Options**
FX Option (§16)
| $F = S \cdot \text{fx_factor} / DF(T)$
| $d1 = [\ln(F/K) + \frac{1}{2} \sigma^2 T] / (\sigma \sqrt{T}); \quad d2 = d1 - \sigma \sqrt{T}$
| $V_{\text{call}} = N \cdot DF(T) \cdot [F \cdot N(d1) - K \cdot N(d2)]$
| $V_{\text{put}} = N \cdot DF(T) \cdot [K \cdot N(-d2) - F \cdot N(-d1)]$
Right to exchange currencies at K on T.
Risk: FX (delta), vol (vega), rate
Swaption (§17)
| $A = \sum \tau \cdot DF(t_i) \quad (\text{swap annuity})$
| $S = [DF(T_{\text{exp}}) - DF(T_{\text{end}})] / A \quad (\text{fwd swap rate})$
| d1, d2 as Black on S
| $V_{\text{payer}} = N \cdot A \cdot [S \cdot N(d1) - K \cdot N(d2)]$
| $V_{\text{recv}} = N \cdot A \cdot [K \cdot N(-d2) - S \cdot N(-d1)]$
Option to enter IRS at K. Black model on forward swap rate.
Risk: rate, vol (vega dominates at ATM)
E11 Cap / Floor / Caplet
| Caplet at t_i : $\tau \cdot N \cdot \max(L_i - K, 0)$ paid at t_{i+1}
| Cap = $\sum$ caplets, each Black on fwd rate
Hedge floating-rate exposure. Pure vol play.
E12 Bermudan Swaption
| Multiple exercise dates → LSMC / lattice
| V > European by "early exercise premium"
Used for callable bond hedging.
Risk: rate, vol, mean reversion (model dep.)
E15 Variance Swap

Payoff = $N \cdot \text{var} \cdot (\sigma^2_{\text{real}} - K^2)$
 $N_{\text{vega}} = 2K \cdot N_{\text{var}}$ (approx near K)
Replication: OTM calls + puts (1/K² weights)
Pure realised-vol exposure; tail hedge.
E16 Equity TRS / E17 Commodity Fwd
| Equity TRS: $V = \text{PV}(\text{equity total return}) - \text{funding}$
| Commodity Fwd: $V = N \cdot (F_T - K) \cdot DF(T)$
Synthetic equity / hedge raw materials.

■ **VII — Liabilities**
Annuity Liability (§18)
| $V = -\text{payment} \cdot \sum (1+b)^{-t} \cdot DF(t), \quad t=1..years$
Pension-style: you OWE inflation-linked payments. NEGATIVE PV.
Risk: rate (rates ↓ → V more neg), breakeven (b ↑ → V more neg).
Hedge: Linker matches IE01 partly; IRS matches DV01.
Sign trap: liability has opposite-sign DV01 to a long bond. Pick one DV01 convention across your stack. KRD-mismatched hedges (parallel-neutral, long-end-exposed) are the canonical pension hedge failure.

◆ **VIII — Risk Metrics**
Macauley Duration (years)
| $D_{\text{mac}} = \sum t_i \cdot PV_i / \sum PV_i$
PV-weighted average time to cashflows.
Modified Duration (%)
| $D_{\text{mod}} = D_{\text{mac}} / (1 + y/m)$
| $\Delta P/P \approx -D_{\text{mod}} \cdot \Delta y$
DV01 / PV01 (\$)
| $DV01 = D_{\text{mod}} \cdot P \cdot 0.0001$
\$ P&L per 1 bp rate move.
Convexity
| $C = \sum t_i (t_i + 1/m) \cdot PV_i / [P \cdot (1 + y/m)^2]$
| $\Delta P/P \approx -D_{\text{mod}} \cdot \Delta y + \frac{1}{2} \cdot C \cdot \Delta y^2$
Positive for long vanilla bonds → cushions losses, boosts gains.
Greeks (Black on forward F)

Δ delta	DF(T) · N(d1)
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Γ gamma	$DF(T) \cdot \phi(d1) / (F \cdot \sigma \cdot \sqrt{T})$
v vega	$DF(T) \cdot F \cdot \phi(d1) \cdot \sqrt{T}$
Θ theta	– for long opt (time decay)
ρ rho	call-rho > 0

Gamma uses **F**, not S, in the denominator.
Numerical (FD) when no closed-form
| $DV01_{fd} = [V(+1bp) - V(-1bp)] / 2$
| $Conv_{fd} = [V(+h) - 2V + V(-h)] / (h^2 \cdot V)$

◆ **Risk-factor sign cheatsheet**

Long position in...	DV01	IE01	FX Δ
Bond / Loan / Linker	–	+ (Linker only)	0
IRS receive-fixed	–	0	0
IRS pay-fixed	+	0	0
ZCIS receive-infl	≈0 at par	+	0
FX Spot / Fwd (long foreign)	+	0	+
Annuity Liability (short)	+	–	0
FX/Swaption (long call)	small	0	+

Convention: signs of $\partial V / \partial r$ (V rises when factor rises = +). Trader DV01 = $-\partial V / \partial r$ just flips signs uniformly. Receive-fix = long duration ⇒ same sign as long bond.

◆ **Top ALM caveats** (full list in gold-ref PDF)

- **Multi-curve:** OIS discount + LIBOR/SOFR projection (drills: single-curve).
- **LDI regimes** (UK FRS102 / US IAS 19): liab discounted on AA-corp → swap-vs-AA basis is unhedged.
- **LPI caps:** UK pension inflation capped at 5% → liability non-linear; ZCIS doesn't hedge convexity.
- **Sept 2022 gilt crisis:** LDI hedge collateral calls cascaded — the hedge itself has liquidity risk.
- **Fisher:** nominal ≈ real + breakeven. Linker DV01 = REAL-rate sensitivity, not nominal.